

# Probability

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The “probability” of an event  $A$  in an experiment is supposed to measure how frequently  $A$  is about to occur if we make many trials.

There are three basic interpretations of probability:

- 1 Classical probability
- 2 Empirical or relative frequency probability
- 3 Subjective probability

Classical probability uses sample spaces to determine the numerical probability that an event will happen. You do not actually have to perform the experiment to determine that probability.<sup>1</sup>

*Classical probability assumes that all outcomes in the sample space are equally likely to occur.*

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<sup>1</sup>it was the first type of probability studied formally by mathematicians in the 17th and 18th centuries.

# Classical probability

## Tossing a coin

If we flip a coin, then heads  $H$  and tails  $T$  will appear about equally often, we say that  $H$  and  $T$  are “equally likely.”

## Tossing a Fair Die

For a regularly shaped die of homogeneous material (“fair die”) each of the six outcomes 1, 2, 3, 4, 5, 6 will be equally likely.

## Selecting Card from Ordinary deck

When a card is selected from an ordinary deck of 52 cards, you assume that the deck has been shuffled, and each card has the same probability of being selected. In this case, it is  $1/52$ .

# Formula for Classical Probability

The probability of any event  $E$  is

$$\frac{\text{Number of outcomes in } E}{\text{Total number of outcomes in the sample space}}$$

This probability is denoted by

$$P(E) = \frac{n(E)}{n(S)}$$

This probability is called classical probability, and it uses the sample space  $S$ .

# Empirical or relative frequency probability

**Empirical probability** The difference between classical and empirical probability is that classical probability assumes that certain outcomes are equally likely (such as the outcomes when a die is rolled), while empirical probability relies on actual experience to determine the likelihood of outcomes. In empirical probability, one might actually roll a given die 6000 times, observe the various frequencies, and use these frequencies to determine the probability of an outcome.

**Formula for Empirical probability** Given a frequency distribution, the probability of an event being in a given class is

$$P(E) = \frac{\text{frequency for the class}}{\text{total frequencies in the distribution}} = \frac{f}{n}$$

This probability is called empirical probability and is based on observation.

### Example

A card is drawn from an ordinary deck. Find these probabilities.

- (a) Of getting a jack
- (b) Of getting the 6 of clubs (i.e., a 6 and a club)
- (c) Of getting a 3 or a diamond
- (d) Of getting a 3 or a 6

$$\begin{aligned} \text{(a)-}P(\text{jack}) &= \frac{4}{52}, \text{ (b)-}P(\text{6 of clubs}) = 1/52, \text{ (c)-}P(\text{3 or diamond}) = \frac{16}{52}, \\ \text{(d)-}P(\text{3 or a 6}) &= \frac{8}{52} \end{aligned}$$

## Example

In a sample of 50 people, 21 had type O blood, 22 had type A blood, 5 had type B blood, and 2 had type AB blood. Set up a frequency distribution and find the following probabilities.

- (i) A person has type O blood.
- (ii) A person has type A or type B blood.
- (iii) A person has neither type A nor type O blood.
- (iv) A person does not have type AB blood

| Type | Frequency |
|------|-----------|
| A    | 22        |
| B    | 5         |
| AB   | 2         |
| O    | 21        |
|      | Total 50  |

$$(i)=22/50, (ii)=27/50, (iii)=7/50, (iv)=1-\frac{2}{50}.$$

# Four Basic Rules for Probability

The probability of any event  $E$  is a number (either a fraction or decimal) between and including 0 and 1. This is denoted by

$$0 \leq P(E) \leq 1.$$

**Rule 1 states that probabilities cannot be negative or greater than 1.**

If an event  $E$  cannot occur (i.e., the event contains no members in the sample space), its probability is 0.

**When a single die is rolled, find the probability of getting a 9.**

If an event  $E$  is certain, then the probability of  $E$  is 1.

**When a single die is rolled, what is the probability of getting a number less than 7?**

The sum of the probabilities of all the outcomes in the sample space is 1.

$$\begin{aligned} 0 \leq P(E) \leq 1, \quad E \subseteq S \\ P(S) = 1, \quad \text{For entire space } S \\ \sum P(E) = 1 \end{aligned} \tag{1}$$

For example, in the roll of a fair die, each outcome in the sample space has a probability of  $\frac{1}{6}$ . Hence, the sum of the probabilities of the outcomes is as shown.

| Outcome     | 1             |   | 2             |   | 3             |   | 4             |   | 5             |   | 6                               |
|-------------|---------------|---|---------------|---|---------------|---|---------------|---|---------------|---|---------------------------------|
| Probability | $\frac{1}{6}$ |   | $\frac{1}{6}$ |   | $\frac{1}{6}$ |   | $\frac{1}{6}$ |   | $\frac{1}{6}$ |   | $\frac{1}{6}$                   |
| Sum         | $\frac{1}{6}$ | + | $\frac{1}{6}$ | + | $\frac{1}{6}$ | + | $\frac{1}{6}$ | + | $\frac{1}{6}$ | + | $\frac{1}{6} = \frac{6}{6} = 1$ |

For example, in the roll of a fair die, each outcome in the sample space has a probability of  $\frac{1}{6}$ . Hence, the sum of the probabilities of the outcomes is as shown.

|                    |               |   |               |   |               |   |               |   |               |   |                                 |
|--------------------|---------------|---|---------------|---|---------------|---|---------------|---|---------------|---|---------------------------------|
| <b>Outcome</b>     | 1             |   | 2             |   | 3             |   | 4             |   | 5             |   | 6                               |
| <b>Probability</b> | $\frac{1}{6}$ |   | $\frac{1}{6}$ |   | $\frac{1}{6}$ |   | $\frac{1}{6}$ |   | $\frac{1}{6}$ |   | $\frac{1}{6}$                   |
| <b>Sum</b>         | $\frac{1}{6}$ | + | $\frac{1}{6}$ | + | $\frac{1}{6}$ | + | $\frac{1}{6}$ | + | $\frac{1}{6}$ | + | $\frac{1}{6} = \frac{6}{6} = 1$ |

$$\begin{aligned}
 0 &\leq P(E) \leq 1, \quad E \subseteq S \\
 P(S) &= 1, \quad \text{For entire space } S \\
 \sum P(E) &= 1
 \end{aligned}$$

Three horses  $A$ ,  $B$  and  $C$  are in a race;  $A$  is twice as likely to win as  $B$  and  $B$  is twice as likely to win as  $C$ . What are their respective probabilities of winning?

Let the probability of winning of horse  $C$  is  $x$ .

then probability of winning  $B$  is  $2x$  and probability of winning  $A$  is  $4x$ .

Also total probability of winning is 1.

$$\text{So } P(A) + P(B) + P(C) = 1$$

$$\Rightarrow x + 2x + 4x = 1 \Rightarrow x = \frac{1}{7}$$

Hence probability of winning of horses  $A$ ,  $B$ , and  $C$  are  $\frac{4}{7}$ ,  $\frac{2}{7}$  and  $\frac{1}{7}$  respectively.

# Complementary Events

Another important concept in probability theory is that of complementary events. When a die is rolled, for instance, the sample space consists of the outcomes 1, 2, 3, 4, 5, and 6. The event  $E$  of getting odd numbers consists of the outcomes 1, 3, and 5. The event of not getting an odd number is called the complement of event  $E$ , and it consists of the outcomes 2, 4, and 6.

## Complement of an event $E$

The *complement of an event  $E$*  is the set of outcomes in the sample space that are not included in the outcomes of event  $E$ . The complement of  $E$  is denoted by  $E^c = S \setminus E$ .

## Rule for Complementary Events

$$P(E^c) = 1 - P(E) \quad E \subseteq S. \quad (2)$$

If the probability of an event or the probability of its complement is known, then the other can be found by subtracting the probability from 1.

Let a coin and a die be tossed; let the sample space  $S$  consist of twelve elements

(i) Express explicitly the following events

$A = \{\text{Heads and even numbers appear}\}$

$B = \{\text{A prime number appears}\}$

$C = \{\text{Tails and odd numbers appear}\}$

(ii) Express explicitly that events

(a)  $A$  or  $B$  occurs

(b)  $B$  and  $C$  occurs

(c) Only  $B$  occurs

(iii) Which of the events  $A$ ,  $B$  and  $C$  are mutually exclusive?

Five coins are tossed simultaneously. Find the probability of the event  $E$ : At least one head turns up. Assume that the coins are fair.

|    |   |   |   |   |   |
|----|---|---|---|---|---|
| 1  | H | H | H | H | H |
| 2  | H | H | H | H | T |
| 3  | H | H | H | T | H |
| 4  | H | H | H | T | T |
| 5  | H | H | T | H | H |
| 6  | H | H | T | H | T |
| 7  | H | H | T | T | H |
| 8  | H | H | T | T | T |
| 9  | H | T | H | H | H |
| 10 | H | T | H | H | T |
| 11 | H | T | H | T | H |
| 12 | H | T | H | T | T |
| 13 | H | T | T | H | H |
| 14 | H | T | T | H | T |
| 15 | H | T | T | T | H |
| 16 | H | T | T | T | T |
| 17 | T | H | H | H | H |
| 18 | T | H | H | H | T |
| 19 | T | H | H | T | H |
| 20 | T | H | H | T | T |
| 21 | T | H | T | H | H |
| 22 | T | H | T | H | T |
| 23 | T | H | T | T | H |
| 24 | T | H | T | T | T |
| 25 | T | T | H | H | H |
| 26 | T | T | H | H | T |
| 27 | T | T | H | T | H |
| 28 | T | T | H | T | T |
| 29 | T | T | T | H | H |
| 30 | T | T | T | H | T |
| 31 | T | T | T | T | H |
| 32 | T | T | T | T | T |

<sup>2</sup>that is TTTTT

Five coins are tossed simultaneously. Find the probability of the event  $E$ : At least one head turns up. Assume that the coins are fair.

|    |   |   |   |   |   |
|----|---|---|---|---|---|
| 1  | H | H | H | H | H |
| 2  | H | H | H | H | T |
| 3  | H | H | H | T | H |
| 4  | H | H | H | T | T |
| 5  | H | H | T | H | H |
| 6  | H | H | T | H | T |
| 7  | H | H | T | T | H |
| 8  | H | H | T | T | T |
| 9  | H | T | H | H | H |
| 10 | H | T | H | H | T |
| 11 | H | T | H | T | H |
| 12 | H | T | H | T | T |
| 13 | H | T | T | H | H |
| 14 | H | T | T | H | T |
| 15 | H | T | T | T | H |
| 16 | H | T | T | T | T |
| 17 | T | H | H | H | H |
| 18 | T | H | H | H | T |
| 19 | T | H | H | T | H |
| 20 | T | H | H | T | T |
| 21 | T | H | T | H | H |
| 22 | T | H | T | H | T |
| 23 | T | H | T | T | H |
| 24 | T | H | T | T | T |
| 25 | T | T | H | H | H |
| 26 | T | T | H | H | T |
| 27 | T | T | H | T | H |
| 28 | T | T | H | T | T |
| 29 | T | T | T | H | H |
| 30 | T | T | T | H | T |
| 31 | T | T | T | T | H |
| 32 | T | T | T | T | T |

Since each coin can turn up heads or tails, the sample space consists of 32 outcomes. Since the coins are fair, we may assign the same probability to each outcome. Then the event (No heads turn up) consists of only 1 outcome<sup>2</sup>. Hence  $P(E^c) = 1/32$ . Thus  $P(E) = 1 - \frac{1}{32} = \frac{31}{32}$ .

<sup>2</sup>that is TTTTT

Hospital records indicated that Dengue patients stayed in the hospital for the number of days shown in the distribution.

| Number of days | Frequency |
|----------------|-----------|
| 3              | 15        |
| 4              | 32        |
| 5              | 56        |
| 6              | 19        |
| 7              | 5         |
|                | 127       |

Find these probabilities.

a. A patient stayed exactly 5 days  $P(5) = \frac{56}{127}$

b. A patient stayed fewer than 6 days  $P(3, 4, 5) = \frac{15+32+56}{127} = \frac{103}{127}$

c. A patient stayed at most 4 days  $P(3, 4) = \frac{15+32}{127} = \frac{47}{127}$

d. A patient stayed at least 5 days  $P(5, 6, 7) = \frac{56+19+5}{127} = \frac{80}{127}$

# The Addition Rules for Probability

Many problems involve finding the probability of two or more events. For example, at a large political gathering, you might wish to know, for a person selected at random, the probability that the person is a female or is a Republican. In this case, there are three possibilities to consider:

1. The person is a female.
2. The person is a Republican.
3. The person is both a female and a Republican.

Consider another example. At the same gathering there are Republicans, Democrats, and Independents. If a person is selected at random, what is the probability that the person is a Democrat or an Independent? In this case, there are only two possibilities:

1. The person is a Democrat.
2. The person is an Independent.

The difference between the two examples is that in the first case, the person selected can be a female and a Republican at the same time. In the second case, the person selected cannot be both a Democrat and an Independent at the same time. In the second case, the two events are said to be *mutually exclusive*; in the first case, they are not mutually exclusive.

# Mutually Exclusive Events

Two events are mutually exclusive events if they cannot occur at the same time (i.e., they have no outcomes in common). *Two events A and B are mutually exclusive events if  $A \cap B = \emptyset$ .*

Determine which events are mutually exclusive and which are not, when a single die is rolled.

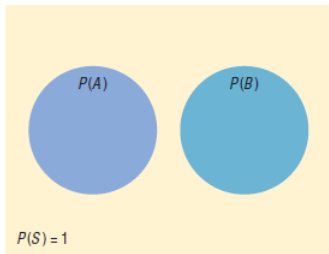
- a. Getting an odd number and getting an even number
- b. Getting a 3 and getting an odd number
- c. Getting an odd number and getting a number less than 4
- d. Getting a number greater than 4 and getting a number less than 4

**Probability for Mutually Exclusive Events** When two events  $A$  and  $B$  are mutually exclusive, the probability that  $A$  or  $B$  will occur is

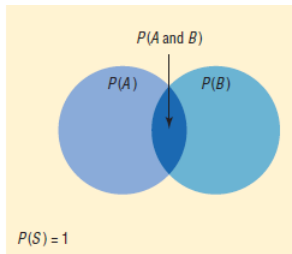
$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) \quad (3)$$

**Probability for non-Mutually Exclusive Events** When two events  $A$  and  $B$  are mutually exclusive, the probability that  $A$  or  $B$  will occur is

$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) - P(A \cap B). \quad (4)$$



(a) Mutually exclusive events  
 $P(A \text{ or } B) = P(A) + P(B)$



(b) Nonmutually exclusive events  
 $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

- 1 A day of the week is selected at random. Find the probability that it is a weekend day.

- ① A day of the week is selected at random. Find the probability that it is a weekend day. **Solution:**  
 $P(\text{Saturday or Sunday}) = 2/7$
- ② A single card is drawn at random from an ordinary deck of cards. Find the probability that it is either a king or a black card.

- ① A day of the week is selected at random. Find the probability that it is a weekend day. **Solution:**  
 $P(\text{Saturday or Sunday}) = 2/7$
- ② A single card is drawn at random from an ordinary deck of cards. Find the probability that it is either an king or a black card. **Solution:**  
 $P(\text{king or black card}) = P(\text{king}) + P(\text{black card}) - P(\text{black king})$

- ① A day of the week is selected at random. Find the probability that it is a weekend day. **Solution:**

$$P(\text{Saturday or Sunday}) = 2/7$$

- ② A single card is drawn at random from an ordinary deck of cards. Find the probability that it is either a king or a black card. **Solution:**

$$P(\text{king or black card}) = P(\text{king}) + P(\text{black card}) - P(\text{black king}) = 4/52 + 26/52 - 2/52 = 7/13.$$

*For three events that are not mutually exclusive,  $P(A \text{ or } B \text{ or } C) =$*

$$\begin{aligned} P(A \cup B \cup C) = & P(A) + P(B) + P(C) \\ & - \{P(A \cap B) + P(B \cap C) + P(C \cap A)\} \\ & + P(A \cap B \cap C) \end{aligned}$$

A class contains 10 men and 20 women of which half the men and half the women have brown eyes. Find the probability  $p$  that a person is chosen at random is a man or has brown eyes.

Let  $P(A)$  be the event that it is a man (10 out of 30)

Let  $P(B)$  be that the person has brown eyes (5 men and 10 woman - 15 out of 30)

(  $P(A \cap B)$  is a man AND has brown eyes 5/30)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= 10/30 + 15/30 - 5/30$$

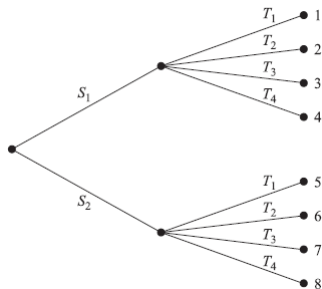
$$= 2/3$$

# Tree Diagram

A tree diagram is a device consisting of line segments emanating from a starting point and also from the outcome point. It is used to determine all possible outcomes of a probability experiment.

## Example

Letting the shirts be represented by  $S_1$ ,  $S_2$  and the ties by  $T_1, T_2, T_3, T_4$ , the various ways of choosing a shirt and then a tie are indicated in the tree diagram of Fig



# The Multiplication Rules and Conditional Probability

# The Multiplication Rules

The multiplication rules can be used to find the probability of two or more events that occur in sequence. For example, if you toss a coin and then roll a die, you can find the probability of getting a head on the coin and a 4 on the die. These two events are said to be independent since the outcome of the first event (tossing a coin) does not affect the probability outcome of the second event (rolling a die).

## Independent Events

Two events A and B are independent events if the fact that A occurs does not affect the probability of B occurring.

Rolling a die and getting a 6, and then rolling a second die and getting a 3.

## Dependent Events

When the outcome or occurrence of the first event affects the outcome or occurrence of the second event in such a way that the probability is changed, the events are said to be dependent events.

Here are some examples of dependent events:

Drawing a card from a deck, not replacing it, and then drawing a second card.

Selecting a ball from an urn, not replacing it, and then selecting a second ball.

### Independent Events (Multiplication Rule 1)

When two events are independent, the probability of both occurring is

$$P(A \text{ and } B) = P(A \cap B) = P(A).P(B) \quad (5)$$

### Dependent Events (Multiplication Rule 2)

When two events are dependent, the probability of both occurring is

$$P(A \text{ and } B) = P(A \cap B) = P(A).P(B/A) \quad P(B/A) = \frac{P(A \cap B)}{P(A)} \quad (6)$$

$P(B/A)$  denotes the probability of  $B$  given that event  $A$  has occurred.

For any three events  $A_1$ ,  $A_2$ ,  $A_3$ , we have

$$P(A_1 \cap A_2 \cap A_3) = P(A_1).P(A_2/A_1).P(A_3/(A_1 \cap A_2))$$

In words, the probability that  $A_1$  and  $A_2$  and  $A_3$  all occur is equal to the probability that  $A_1$  occurs times the probability that  $A_2$  occurs given that  $A_1$  has occurred times the probability that  $A_3$  occurs given that both  $A_1$  and  $A_2$  have occurred.

Three cards are drawn from an ordinary deck and not replaced. Find the probability of these events.

- 1 Getting 3 jacks
- 2 Getting an ace a king, and a queen in order
- 3 Getting a club, a spade, and a heart in order
- 4 Getting 3 clubs

Three cards are drawn from an ordinary deck and not replaced. Find the probability of these events.

- 1 Getting 3 jacks
- 2 Getting an ace a king, and a queen in order
- 3 Getting a club, a spade, and a heart in order
- 4 Getting 3 clubs

**Solution:**

$$(a) P(3 \text{ jacks}) = \frac{4}{52} \cdot \frac{3}{51} \cdot \frac{2}{50} = \frac{1}{5525}$$

$$(b) P(\text{ace and king and queen}) = \frac{4}{52} \cdot \frac{4}{51} \cdot \frac{4}{50}$$

$$(c) P(\text{club and spade and heart}) = \frac{13}{52} \cdot \frac{13}{51} \cdot \frac{13}{50}$$

$$(d) P(3 \text{ clubs}) = \frac{13}{52} \cdot \frac{12}{51} \cdot \frac{11}{50}$$

Two cards are drawn from a well-shuffled ordinary deck of 52 cards.

Find the probability that they are both aces if the first card is (a) replaced, (b) not replaced.

There are 52 cards in a deck, and there are 4 aces in a deck, so the probability of drawing an ace is  $\frac{4}{52}$

If you do not replace the card, there are only 51 cards left and only 3 aces left, so the probability of now drawing an ace is  $\frac{3}{51}$

Multiply these probabilities:

$$\frac{4}{52} \times \frac{3}{51} = \frac{12}{2652} \rightarrow 0.00452$$

- 1 Find the probability that a single toss of a die will result in a number less than 4 if
- (a) no other information is given
  - (b) it is given that the toss resulted in an odd number.

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**Solution:**

$$B = \{\text{No. less than 4}\}, \quad A = \{\text{Odd No.}\}, \quad P(B) = \frac{1}{2},$$

$$P(A) = ?, \quad P(B) = ?, \quad P(A \cap B) = ?, \quad \boxed{P(B/A) = ?}$$

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What do you mean by  $P(B/A)$  and  $P(A \cap B)$ ?

***The notation  $B/A$  means  $B$  occurs after event  $A$  has already occurred.***

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- (a) no other information is given
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**Solution:**

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What do you mean by  $P(B/A)$  and  $P(A \cap B)$ ?

- 2 World Wide Insurance Company found that 53% of the residents of a city had home owner's insurance (H) with the company. Of these clients, 27% also had automobile insurance (A) with the company. If a resident is selected at random, find the probability that the resident has both home owner's and automobile insurance with World Wide Insurance Company.

- ① Find the probability that a single toss of a die will result in a number less than 4 if
- (a) no other information is given
  - (b) it is given that the toss resulted in an odd number.

**Solution:**

$$B = \{\text{No. less than 4}\}, \quad A = \{\text{Odd No.}\}, \quad P(B) = \frac{1}{2},$$

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What do you mean by  $P(B/A)$  and  $P(A \cap B)$ ?

- ② World Wide Insurance Company found that 53% of the residents of a city had home owner's insurance (H) with the company. Of these clients, 27% also had automobile insurance (A) with the company. If a resident is selected at random, find the probability that the resident has both home owner's and automobile insurance with World Wide Insurance Company.

$$P(H) = 0.53, \quad P(A/H) = 0.27, \quad P(H \cap A) = ?$$

- ① A fair die is tossed twice. Find the probability of getting a 4, 5, or 6 on the first toss and a 1, 2, 3, or 4 on the second toss.

Let  $A_1$  be the event “4, 5, or 6 on first toss,” and  $A_2$  be the event “1, 2, 3, or 4 on second toss.” Then we are looking for  $P(A_1 \cap A_2)$ .

**Method#1** Each of the 3 ways in which  $A_1$  can occur can be associated with each of the 4 ways in which  $A_2$  can occur to give  $3 \cdot 4 = 12$  ways in which both  $A_1$  and  $A_2$  can occur out of total 36 ways. Then

$$P(A_1 \cap A_2) = \frac{12}{36}$$

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**Method#2**

$$P(A_1 \cap A_2) = P(A_1)P(A_2/A_1)$$

The fact that the result of the second toss is independent of the first so that  $P(A_2/A_1) = P(A_2)$ . Thus

$$P(A_1 \cap A_2) = P(A_1)P(A_2) =$$

Suppose that our sample space  $S$  is the population of adults in a small town who have completed the requirements for a college degree. We shall categorize them according to gender and employment status. The data are given in Table

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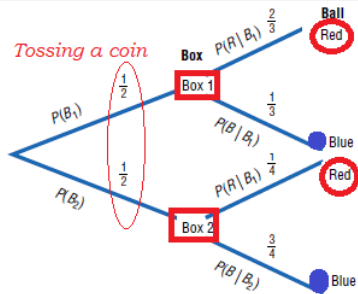
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Suppose  $M$  : a male is chosen and  $E$  : the chosen one is employed. Find

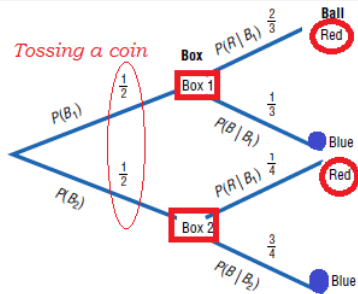
$$P(M/E)$$

Box-1 contains 2 red balls and 1 blue ball. Box-2 contains 3 blue balls and 1 red ball. A coin is tossed. If it falls heads up, Box-1 is selected and a ball is drawn. If it falls tails up, Box-2 is selected and a ball is drawn. Find the probability of selecting a red ball.

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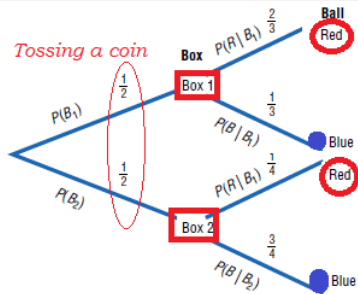
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- 2 Find red ball.

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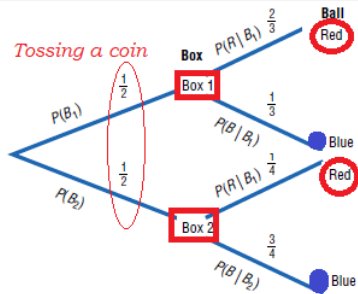


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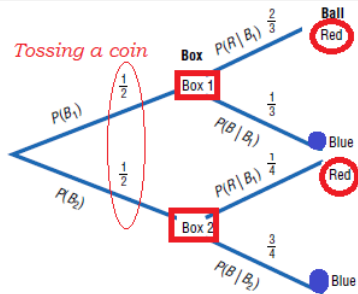
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$$P(\text{red ball}) = P(B1)P(R/B1) + P(B2)P(R/B2) = \frac{2}{6} + \frac{1}{8} = \frac{11}{24}$$

Next determine the probabilities for each branch. Since a coin is being tossed for the box selection, each branch has a probability of  $\frac{1}{2}$  that is, heads for box 1 or tails for box 2. The probabilities for the second branches are found by using the basic probability rule. For example, if box 1 is selected and there are 2 red balls and 1 blue ball, the probability of selecting a red ball is  $\frac{2}{3}$  and the probability of selecting a blue ball is  $\frac{1}{3}$ . If box 2 is selected and it contains 3 blue balls and 1 red ball, then the probability of selecting a red ball is  $\frac{1}{4}$  and the probability of selecting a blue ball is  $\frac{3}{4}$ .

Next multiply the probability for each outcome, using the rule

$$P(\text{red and Box} - 1) = P(\text{Box} - 1) \cdot P(\text{red} / \text{Box} - 1).$$

For example, the probability of selecting box 1 and selecting a red ball is

$$\frac{1}{2} \cdot \frac{2}{3} = \frac{2}{6}.$$

*The probability of selecting box 2 and selecting a red ball is obtained using the rule  $P(\text{red and Box} - 2) = P(\text{Box} - 2) \cdot P(\text{red} / \text{Box} - 2)$ .*

$$\frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}.$$

The probability of selecting box 2 and a blue ball is (Note that the sum of these probabilities is 1.)

*Finally a red ball can be selected from either box 1 or box 2 so*

$$P(\text{red}) = \frac{2}{6} + \frac{1}{8} = \frac{11}{24}.$$

A recent survey asked 100 people if they thought women in the armed forces should be permitted to participate in combat. The results of the

survey are shown.

| Gender | Yes | No | Total |
|--------|-----|----|-------|
| Male   | 32  | 18 | 50    |
| Female | 8   | 42 | 50    |
| Total  | 40  | 60 | 100   |

Find these probabilities.

- The respondent answered yes, given that the respondent was a female.
- The respondent was a male, given that the respondent answered no.

**Solution:**

Let

M = respondent was a male  
F = respondent was a female

Y = respondent answered yes  
N = respondent answered no

| Gender | Yes | No | Total |
|--------|-----|----|-------|
| Male   | 32  | 18 | 50    |
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$$P(Y/F) = \frac{P(F \text{ and } Y)}{P(F)}$$

The probability  $P(F \text{ and } Y)$  is the number of females who responded yes, divided by the total number of respondents: Thus  $P(F \text{ and } Y) = 8/100$ . The probability  $P(F)$  is the probability of selecting a female:  $P(F) = \frac{50}{100}$ .

$$P(Y/F) = \frac{P(F \text{ and } Y)}{P(F)} = \frac{8/100}{50/100} = \frac{4}{25}$$

The problem is to find  $P(Y/F)$ . The rule states b. The problem is to find  $P(M/N)$ .

$$P(M/N) = \frac{P(M \text{ and } N)}{P(N)} = \frac{18/100}{60/100} = \frac{3}{10}$$

- ① A fair die is tossed twice. Find the probability of getting a 4, 5, or 6 on the first toss and a 1, 2, 3, or 4 on the second toss.
- ② The probability that Sam parks in a no-parking zone and gets a parking ticket is 0.06, and the probability that Sam cannot find a legal parking space and has to park in the no parking zone is 0.20. On Tuesday, Sam arrives at school and has to park in a no-parking zone. Find the probability that he will get a parking ticket.

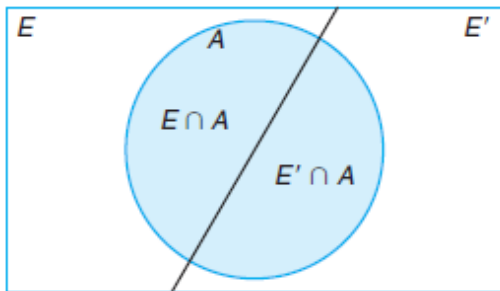
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- 1 Suppose that we are now given the additional information that 36 of those employed and 12 of those unemployed are members of the Rotary Club.
- 2 We wish to find the probability of the event  $A$  that the individual selected is a member of the Rotary Club.

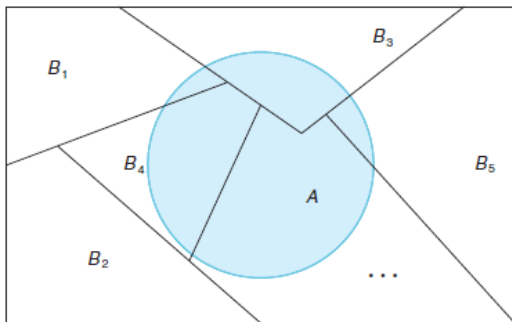
A is union of two mutually exclusive events  $E \cap A$  and  $E^c \cap A$

$$\begin{aligned} P(A) &= P[(E \cap A) \cup (E^c \cap A)] = P(E \cap A) + P(E^c \cap A) \\ &= P(E)P(A/E) + P(E^c)P(A/E^c) \end{aligned}$$



If the events  $B_1, B_2, \dots, B_k$  constitute a partition of the sample space  $S$  such that  $P(B_i) \neq 0$  for  $i = 1, 2, \dots, k$ , then for any event  $A$  of  $S$ ,

$$P(A) = \sum_{i=1}^k P(B_i \cap A) = \sum_{i=1}^k P(B_i)P(A/B_i).$$



# Bayes' Theorem or Rule

In a certain assembly plant, three machines,  $B_1$ ,  $B_2$ , and  $B_3$ , make 30%, 45%, and 25%, respectively, of the products. It is known from past experience that 2%, 3%, and 2% of the products made by each machine, respectively, are defective. Now, suppose that a finished product is randomly selected.

- ① What is the probability that it is defective?
- ② If a product was chosen randomly and found to be defective, what is the probability that it was made by machine  $B_3$ ?

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- 
- ①  $P(\text{the product is defective})$
  - ②  $P(?)$  In this case, the outcome is known, the product is defective, and you are asked to find the probability that it came from  $B_3$

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- 1  $P(A) = P(B_1)P(A/B_1) + P(B_2)P(A/B_2) + P(B_3)P(A/B_3)$
- 2 Find  $P(B_3/A)$ ?

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- 2 Find  $P(B_3/A)$ ?

$$P(B_3/A) = \frac{P(B_3 \cap A)}{P(A)} = \frac{P(B_3)P(A/B_3)}{P(B_1)P(A/B_1) + P(B_2)P(A/B_2) + P(B_3)P(A/B_3)}$$

# Bayes' Theorem or Rule

## Theorem

Suppose that  $A_1, A_2, \dots, A_n$  are **mutually exclusive** events whose union is the sample space  $S$ , i.e., one of the events must occur. Then if  $A$  is any event, we have

$$P(A_k/A) = \frac{P(A_k)P(A/A_k)}{\sum_{j=1}^n P(A_j)P(A/A_j)}$$

For two events  $A$  and  $B$ , where event  $B$  follows event  $A$ , event  $A$  can occur in  $A_1, A_2, \dots, A_n$  mutually exclusive ways, and event  $B$  can occur in  $B_1, B_2, \dots, B_m$  mutually exclusive ways, for any specific events  $A_1$  and  $B_1$ .

$$P(A_1/B_1) = \frac{P(A_1).P(B_1/A_1)}{P(A_1).P(B_1/A_1) + P(A_2).P(B_1/A_2) + \dots + P(A_n).P(B_1/A_n)}$$

In the a previous example, there were two boxes, each containing red balls and blue balls. A box was selected and a ball was drawn. The example asked for the probability that the ball selected was red.

$$P(\text{red ball}) = P(B1)P(R/B1) + P(B2)P(R/B2) = \frac{2}{6} + \frac{1}{8} = \frac{11}{24}$$

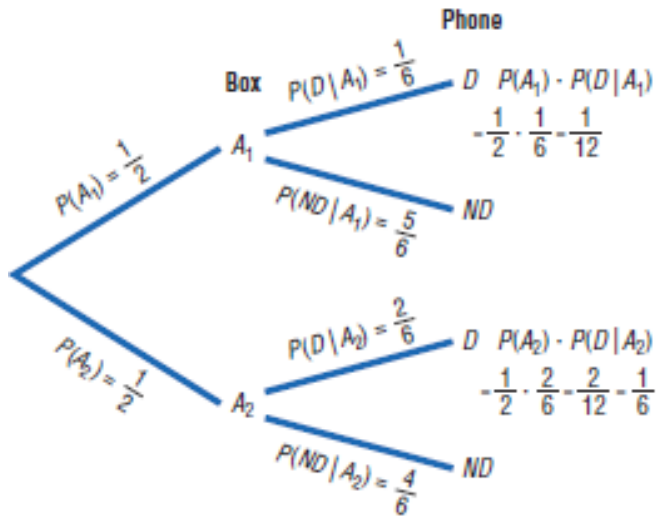
Now, a different question can be asked:

If the ball is red, what is the probability it came from box 1? In this case, the outcome is known, a red ball was selected, and you are asked to find the probability that it is a result of a previous event, that it came from box 1.

$$P(\text{Box1 given red ball}) = P(B1/R) = \frac{P(B1 \cap R)}{P(R)}$$

A shipment of two boxes, each containing six telephones, is received by a store. Box 1 contains one defective phone, and box 2 contains two defective phones. After the boxes are unpacked, a phone is selected and found to be defective. Find the probability that it came from box 2.

- 1 Select the proper notation. Let  $A_1$  represent box 1 and  $A_2$  represent box 2. Let  $D$  represent a defective phone and  $ND$  represent a phone that is not defective.
- 2 Draw a tree diagram and find the corresponding probabilities for each branch. The probability of selecting box 1 is  $\frac{1}{2}$ , and the probability of selecting box 2 is  $\frac{1}{2}$ . Since there is one defective phone in box 1  $\frac{1}{6}$ , the probability of selecting it is  $\frac{1}{6}$ . The probability of selecting a nondefective phone from box 1 is  $\frac{5}{6}$ . Since there are two defective phones in box 2, the probability of selecting a defective phone from box 2 is  $\frac{2}{6}$ , or  $\frac{1}{3}$ ; and the probability of selecting a non defective phone is  $\frac{4}{6}$ , or  $\frac{2}{3}$ . The tree diagram is



Write the corresponding formula. Since the example is asking for the probability that, given a defective phone, it came from box 2, the corresponding formula is as shown.  $P(A_2/D) = ?$

On a game show, a contestant can select one of four boxes. Box 1 contains one \$100 bill and nine \$1 bills. Box 2 contains two \$100 bills and eight \$1 bills. Box 3 contains three \$100 bills and seven \$1 bills. Box 4 contains five \$100 bills and five \$1 bills. The contestant selects a box at random and selects a bill from the box at random. If a \$100 bill is selected, find the probability that it came from box 4.

A real estate agent has 8 master keys to open several new homes. Only 1 master key will open any given house. If 40% of these homes are usually left unlocked, what is the probability that the real estate agent can get into a specific home if the agent selects 3 master keys at random before leaving the office?

The real estate agent can get into a specific home in following conditions

(1)  $\underbrace{\text{The home is unlocked.}}_U$

(2)  $\underbrace{\text{The home is locked}}_L \text{ but } \underbrace{\text{he chose an appropriate key.}}_K$

$$P \left[ \begin{array}{c} \text{Estate agent} \\ \text{can get into} \\ \text{a specific} \\ \text{home} \end{array} \right] = \underbrace{P(U)}_{0.4} + \underbrace{P(L \cap K)}_{P(L) \cdot P(K)},$$

$$P(K) = \frac{(\text{selecting 1 appropriate key})(\text{Selecting 2 keys out of 7})}{\text{choosing 3 out of 8 keys}}$$

A town has two fire engines operating independently. The probability that a specific engine is available when needed is 0.96.

- 1 What is the probability that neither is available when needed?
- 2 What is the probability that a fire engine is available when needed?

**Solution:**

1

$$P(A_1^c \cap A_2^c) = P(A_1^c) \cdot P(A_2^c) = \{1 - P(A_1)\} \cdot \{1 - P(A_2)\}$$

2

P(at least one fire engine is available )

A producer of a certain type of electronic component ships to suppliers in lots of twenty. Suppose that 60% of all such lots contain no defective components, 30% contain one defective component, and 10% contain two defective components. A lot is picked, two components from the lot are randomly selected and tested, and neither is defective.

- 1 What is the probability that zero defective components exist in the lot?
- 2 What is the probability that one defective exists in the lot?
- 3 What is the probability that two defectives exist in the lot?

Let  $D_i$  denote lot contain  $i$  defective components

$$P(D_0) = 0.6, \quad P(D_1) = 0.3, \quad P(D_2) = 0.1$$

$N = \{\text{A lot is picked, two components from the lot are randomly selected and tested, and neither is defective.}\}$   $P(D_0/N) = ?$ ,  $P(D_1/N) = ?$ ,  $P(D_2/N) = ?$

- The probability that 2 nondefective components are chosen from a lot with 0 defectives is:

$$P(N|D_0) = \frac{\begin{array}{c} \text{the number of ways to} \\ \text{choose 2 out of 20} \\ \text{nondefective components} \end{array}}{\begin{array}{c} \text{the number of ways to} \\ \text{choose 2 out of 20} \\ \text{components} \end{array}} = \frac{\binom{20}{2}}{\binom{20}{2}} = 1 \quad (2)$$

- The probability that 2 nondefective components are chosen from a lot with 1 defectives is:

$$P(N|D_1) = \frac{\begin{array}{c} \text{the number of ways to} \\ \text{choose 2 out of 19} \\ \text{nondefective components} \end{array}}{\begin{array}{c} \text{the number of ways to} \\ \text{choose 2 out of 20} \\ \text{components} \end{array}} = \frac{\binom{19}{2}}{\binom{20}{2}} = \frac{171}{190} = 0.9 \quad (3)$$

•

$$P(N|D_2) = \frac{\begin{array}{c} \text{the number of ways to} \\ \text{choose 2 out of 18} \\ \text{nondefective components} \end{array}}{\begin{array}{c} \text{the number of ways to} \\ \text{choose 2 out of 20} \end{array}} = \frac{\binom{18}{2}}{\binom{20}{2}} = \frac{153}{190} \approx 0.8 \quad (4)$$

$$\begin{aligned}
 P(D_0|N) &= \frac{P(N|D_0)P(D_0)}{P(N|D_0)P(D_0) + P(N|D_1)P(D_1) + P(N|D_2)P(D_2)} \\
 &= \frac{1 \cdot 0.6}{1 \cdot 0.6 + 0.9 \cdot 0.3 + 0.8 \cdot 0.1} \\
 &= \frac{0.6}{0.95} \\
 &\approx \mathbf{0.63}
 \end{aligned}$$

$$\begin{aligned}
 P(D_1|N) &= \frac{P(N|D_1)P(D_1)}{P(N|D_0)P(D_0) + P(N|D_1)P(D_1) + P(N|D_2)P(D_2)} \\
 &= \frac{0.9 \cdot 0.3}{1 \cdot 0.6 + 0.9 \cdot 0.3 + 0.8 \cdot 0.1} \\
 &= \frac{0.27}{0.95} \\
 &\approx \mathbf{0.284}
 \end{aligned}$$

$$\begin{aligned}
 P(D_2|N) &= \frac{P(N|D_2)P(D_2)}{P(N|D_0)P(D_0) + P(N|D_1)P(D_1) + P(N|D_2)P(D_2)} \\
 &= \frac{0.8 \cdot 0.1}{1 \cdot 0.6 + 0.9 \cdot 0.3 + 0.8 \cdot 0.1} \\
 &= \frac{0.08}{0.95} \\
 &\approx \mathbf{0.084}
 \end{aligned}$$

A certain form of cancer is known to be found in women over 60 with probability 0.07. A blood test exists for the detection of the disease, but the test is not infallible. In fact, it is known that 10% of the time the test gives a false negative (i.e., the test incorrectly gives a negative result) and 5% of the time the test gives a false positive (i.e., incorrectly gives a positive result). If a woman over 60 is known to have taken the test and received a favorable (i.e., negative) result, what is the probability that she has the disease?

- Let  $C$  be the event that the women over 60 has certain form of cancer  $P(C) = 0.07$ .
- Let  $A$  denotes that test results in positive.
- It is known that 10% of the time the test gives a false negative  $P(A^c/C) = 0.1$
- It is known that 5% of the time the test gives a false positive  $P(A/C^c) = 0.05$ .

$$P(C/A^c) = ?$$

$$P(C/A^c) = \frac{P(A^c/C)P(C)}{P(A^c/C^c)P(C^c) + P(A^c/C)P(C)}$$

$$P(A^c/C^c) = 1 - P(A/C^c) = 1 - .005 = 0.95$$

$$\begin{aligned}
 &= \frac{0.1 \cdot 0.07}{0.95 \cdot 0.93 + 0.1 \cdot 0.07} \\
 &= \frac{0.007}{0.8835 + 0.007} \\
 &= \frac{0.007}{0.8905} \approx 0.0079
 \end{aligned}$$

# QUIZ #1 & QUIZ # 2

- ① An urn contains 5 red balls and 4 green balls. A ball is drawn.
  - (a) If it's green a two red balls are added to the urn and if it's red a one green ball and one red ball are add from the urn. (The original ball is not returned to the urn.) Then a second ball is drawn. What is the probability the second ball is red?
  - (b) If it's green a red ball is added to the urn and if it's red a green ball is added to the urn. (The original ball is not returned to the urn.) Then a second ball is drawn. What is the probability the that both balls are of different colors?
- ② For married couples living in a certain suburb, the probability that the husband will vote on a bond referendum is 0.21, the probability that the wife will vote on the referendum is 0.28, and the probability that both the husband and the wife will vote is 0.15. What is the probability that
  - (i) at least one member of a married couple will vote?
  - (ii) a wife will vote, given that her husband will vote?
  - (iii) a husband will vote, given that his wife will not vote?